

● EASY

● ALGEBRA

Linear equations in one variable

05

10 questions · ~15 min

Q1**ALGEBRA**

If $3x - 8 = 7$, what is the value of $3x + 8$?

- A. -1
- B. 5
- C. 13
- D. 23

The perimeter of an isosceles triangle is 36 feet. Each of the two congruent sides of the triangle has a length of 10 feet. What is the length, in feet, of the third side?

- A. 10
- B. 12
- C. 16
- D. 18

If $\frac{1}{2}x - \frac{1}{6}x = 1$, what is the value of x ?

A. -4

B. $\frac{1}{3}$

C. 3

D. 6

What is the solution to the equation $2x + 3 = 7$?

- A. 1
- B. 1.5
- C. 2
- D. 4

Q5

GRID-IN ALGEBRA

If $2x = 12$, what is the value of $9x$?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

If $7x = 28$, what is the value of $8x$?

- A. 21
- B. 32
- C. 168
- D. 224

$$72x - 3 = 63$$

Which equation has the same solution as the given equation?

A. $2x - 3 = 9$

B. $2x - 3 = 56$

C. $2x - 21 = 63$

D. $2x - 21 = 70$

Henry receives a \$ 60.00 gift card to pay for movies online. He uses his gift card to buy 3 movies for \$ 7.50 each. If he spends the rest of his gift card balance on renting movies for \$ 1.50 each, how many movies can Henry rent?

- A. 10
- B. 25
- C. 35
- D. 40

Q9

GRID-IN ALGEBRA

$$13x = 112 - x$$

What value of x is the solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$k + 12 = 336$$

What is the solution to the given equation?

- A. 28
- B. 324
- C. 348
- D. 4,032